Agentic AI for Robust and Socially Aware Autonomous Driving

Multi-Agent Negotiation for Right-of-Way in Complex Intersections

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Project Proposal Report

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DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

Unsignalized intersections remain one of the most challenging environments for autonomous driving systems. Studies show that approximately 40% of crashes involving human-driven vehicles occur at intersections, while nearly 88% of reported autonomous vehicle crashes happen in these locations. The absence of traffic signals requires vehicles to determine right-of-way through complex interactions with other road users, making safe decision-making difficult for current autonomous systems. Existing solutions such as rule-based coordination, centralized traffic controllers, and game-theoretic approaches suffer from several limitations including lack of scalability, single points of failure, reliance on external safety rules, and inability to support continuous negotiation between vehicles.

This research addresses this gap by proposing a **decentralized multi-agent negotiation framework** for autonomous vehicles at unsignalized intersections. In the proposed system, each vehicle operates as an independent intelligent agent capable of communicating a **priority intent signal** ranging from 0 to 1 that represents its negotiation intention. Using **Multi-Agent Reinforcement Learning (MARL)** with the **Multi-Agent Proximal Policy Optimization (MAPPO)** algorithm, vehicles learn cooperative strategies for negotiating right-of-way through repeated interactions in simulation environments. An **LSTM-based observation encoder** is used to process dynamic neighboring vehicle information, while **Vehicle-to-Vehicle (V2V) communication** enables continuous exchange of negotiation signals.

The system will be implemented and trained using **SUMO and CARLA simulators**, with evaluation conducted using real-world datasets including **INTERACTION, NGSIM, and HighD**. Performance will be assessed using safety and efficiency metrics such as collision rate, travel time, throughput, and negotiation fairness.

The expected outcome is a scalable and robust intersection management system capable of reducing collisions and improving traffic efficiency without centralized control. The proposed framework has potential social and commercial impact by improving road safety, enabling reliable autonomous vehicle deployment, and reducing traffic congestion.

Keywords: Autonomous Vehicles, Multi-Agent Reinforcement Learning, Intersection Management, V2V Communication, Negotiation-Based Coordination.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AV	Autonomous Vehicle
CAV	Connected Autonomous Vehicle
CTDE	Centralized Training with Decentralized Execution
Dec-POMDP	Decentralized Partially Observable Markov Decision Process
DMV	Department of Motor Vehicles
DRL	Deep Reinforcement Learning
FCFS	First-Come First-Served
FSD	Full Self-Driving
HD	High-Definition
HDV	Human-Driven Vehicle
HighD	Highways Drone Dataset
INTERACTION	INTERACTION Dataset (International intersection dataset)
LSTM	Long Short-Term Memory
MADRL	Multi-Agent Deep Reinforcement Learning
MAPPO	Multi-Agent Proximal Policy Optimization
MARL	Multi-Agent Reinforcement Learning
NGSIM	Next Generation Simulation
PET	Post-Encroachment Time
POMDP	Partially Observable Markov Decision Process
RAIM	Reinforced Autonomous Intersection Management
RARL	Risk-Aware Reinforcement Learning
RL	Reinforcement Learning
RWC	Right-of-Way Cooperators
SUMO	Simulation of Urban MObility
TD3	Twin Delayed Deep Deterministic Policy Gradient
TTC	Time-to-Collision
V2V	Vehicle-to-Vehicle Communication

VRCS	Vehicle-Road Cooperative System
WBS	Work Breakdown Structure
XAI	Explainable Artificial Intelligence

1. INTRODUCTION

1.1 Background and Context

Imagine four cars arriving at an intersection with no traffic lights at the same time. How do they know who should go first? This simple question is one of the hardest problems in autonomous driving.



Figure 1 -Example of Multiple Vehicles Arriving Simultaneously at an Unsignalized Intersection

The Real-World Problem

Intersections are one of the most dangerous areas in road traffic. Studies show that around 40% of crashes involving human-driven vehicles occur at intersections, and 50% of serious collisions happen in these locations [1]. The situation is even more challenging for autonomous vehicles. Data from the California Department of Motor Vehicles (DMV) indicates that about 88% of autonomous vehicle crashes occur at intersections, and their crash rate per mile can be 13 to 100 times higher than human-driven vehicles [1]. A 2025 study analyzing 555 autonomous vehicle crashes and 39,270 human-driven crashes found that AVs are nearly twice as likely to crash at intersections (68.5%) compared to human drivers (34.9%) [2]. These findings show that intersections are a critical challenge for autonomous driving systems, especially in unsignalized intersections where vehicles must negotiate right-of-way without traffic signals.

Metric	**Human-Driven Vehicles**	**Autonomous Vehicles**
Crashes at intersections	40% of all crashes	88% of reported crashes
Intersection crash rate	34.9% of crashes	68.5% of crashes
Performance	Baseline	Nearly 2× worse at intersections

Table 1 - Human vs Autonomous Vehicle Crash Comparison [1,2]

Affected Stakeholders

Intersection safety problems affect many groups of people, not just drivers.

Stakeholder	Impact
Passengers in autonomous vehicles	Expect safe and reliable travel without crashes at intersections.
Human drivers	Must safely interact with autonomous vehicles at shared intersections.
Pedestrians and cyclists	Highly vulnerable to accidents at intersections.
Insurance companies	Responsible for covering financial losses caused by crashes.
City transportation authorities	Need to manage traffic flow and reduce congestion at intersections.
Autonomous vehicle manufacturers	Companies such as Tesla, Waymo, and Cruise must improve autonomous vehicle decision-making systems to reduce crashes.

Table 2 - Affected stakeholders and Impact

Statistical Evidence

Statistical evidence shows that intersections are dangerous for all vehicles, but especially for autonomous vehicles.

For Human-Driven Vehicles:

- 40% of all crashes happen at intersections [1]
- 50% of serious collisions occur at intersections [1]
- 20% of traffic fatalities are intersection-related [1]

For Autonomous Vehicles (Even Worse):

- 88% of all autonomous vehicle crashes reported to the California DMV occur at intersections [1]
- Autonomous vehicle crash rate per mile is 13 to 100 times worse than human-driven cars [1]
- A 2025 study of 555 AV crashes and 39,270 human crashes found: autonomous vehicles are nearly twice as likely to crash at intersections compared to humans (68.5% vs. 34.9%) [7]

Why this matters:
If we can solve the intersection problem, we eliminate:

- 40% of human-driven crashes
- 88% of autonomous vehicle crashes

Economic Impact:
Beyond lives lost, intersection congestion costs the U.S. trucking industry $74.1 billion annually in delays and fuel waste [2]. Every second spent waiting at an intersection costs money, fuel, and time.

Analysis of Existing Solutions and Their Limitations

Current approaches to intersection management fall into several categories, each with major limitations:

Approach	What It Does	Limitation
Rule-based (FCFS)	First-come first-served	Cannot handle multiple cars arriving at same time [2]
Centralized MARL	One computer controls all cars	Single point of failure, not realistic [3]
MARL + Rules	Learning with safety rules added	Safety depends on rules, not learning [6]
Game Theory	One-time bids to decide who goes	Real negotiation is continuous, not one decision [4]
Risk-aware RL	Focus on safety modeling	No explicit negotiation between vehicles [5]
Pedestrian-focused	Addresses pedestrian safety	Does not solve vehicle-vehicle negotiation [12]

Table 3 - Analysis of Existing Solutions and Their Limitations

Ideas for the Proposed Solution

This research proposes a fundamental shift from rule-based coordination to emergent, learning-based negotiation. Instead of relying on fixed traffic rules like First-Come First-Served (FCFS) which cannot adapt to complex scenarios [2] or using a central controller that creates a single point of failure [3][10], this system treats each vehicle as an independent, intelligent agent capable of real-time negotiation. The core innovation is an explicit **priority intent** signal a continuous value from 0 to 1 broadcast by every vehicle every 0.1 seconds. This signal acts as a "politeness meter," with 0 indicating yielding, 1 indicating asserting, and intermediate values representing varying assertiveness levels. This builds directly on recent advances in driving style recognition [9], which confirm that driver behavior exists on a spectrum from conservative to aggressive. By sharing these signals through V2V communication, vehicles create a continuous, multi-agent dialogue directly addressing the limitation of one-time bidding systems [4]. The system employs Multi-Agent Proximal Policy Optimization (MAPPO), which has been proven effective for intersection coordination [10][6]. Through trial-and-error learning in the SUMO simulator, agents practice millions of intersection crossings, receiving positive rewards for safe passage and substantial penalties for collisions, a proven method for teaching complex behaviors [3][5][12]. This process allows vehicles to discover sophisticated cooperative strategies yielding to assertive agents, taking turns when all are neutral that are not explicitly programmed but emerge from learning. Critically, this directly tackles the fundamental gap identified by Feng et al.: that pure MARL requires external rules to ensure safety [6]. By making negotiation an integral part of the learning process through explicit signals, this research aims to eliminate the need for those external rules entirely.

1.2 Literature Review

Autonomous Driving at Unsignalized Intersections: A Review of Decision-Making Challenges and Reinforcement Learning-Based Solutions

This comprehensive review paper examines decision-making challenges at unsignalized intersections. The author's state: "making decisions at unsignalized intersections is a highly intractable process" [1]. They review various approaches including Partially Observable Markov Decision Process (POMDP) methods, deep reinforcement learning, and intention inference techniques. The paper identifies that existing methods struggle with scalability and real-world deployment. They also document intersection crash statistics for both human and autonomous vehicles, citing California Department of Motor Vehicles (DMV) data showing that 88% of autonomous vehicles crashes occur at intersections. The paper reviews Long Short-Term Memory (LSTM)-based intention prediction achieving "90% accuracy at intersections with different types and shapes."

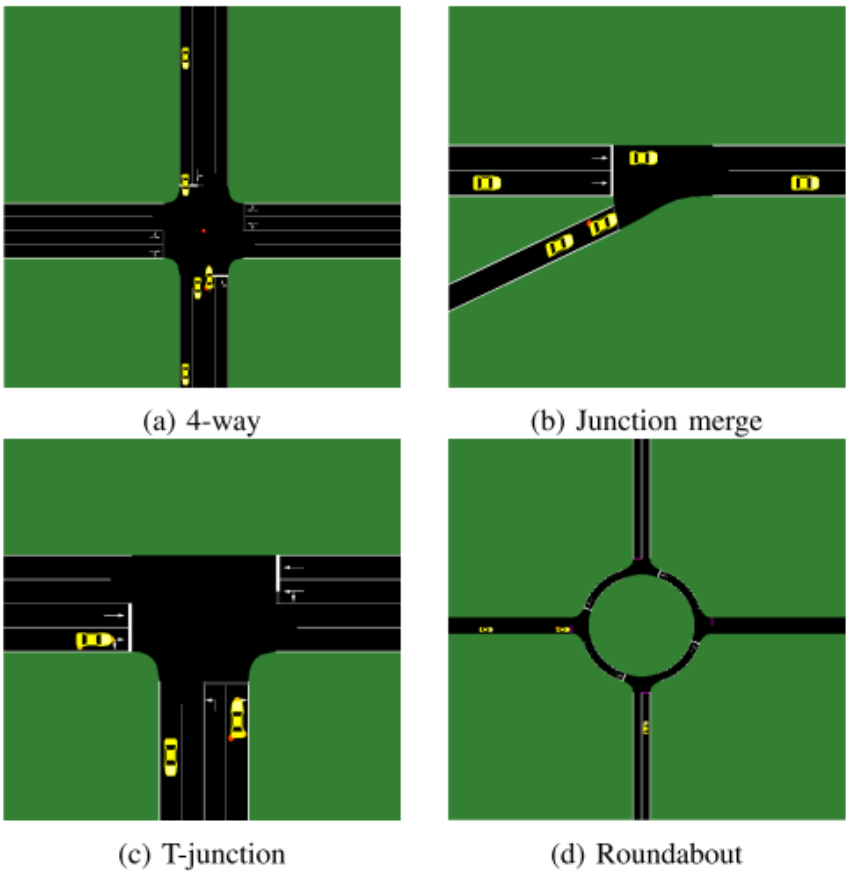


Figure 2 - shows different types of unsignalized intersections including T-junctions, four-way stops, and roundabouts

Cyber Traffic Light: Safe Cooperation for Autonomous Vehicles at Dynamic Intersections

This paper introduces the concept of "Dynamic Intersections" and proposes a cooperative protocol using Cyber Traffic Light - a V2V communication-based traffic manager created by vehicles when congestion arises [2]. The system "allocates the green period for vehicles coming from the multiple directions." While innovative, this creates a temporary central manager rather than true peer-to-peer negotiation. They report traffic efficiency improvements but note that rule-based coordination cannot handle all scenarios.

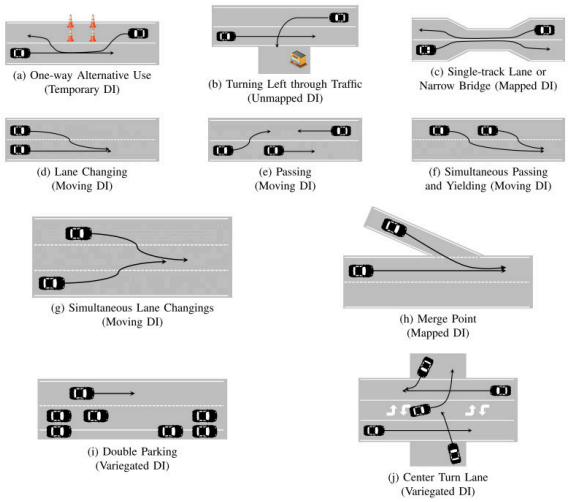


Figure 3 - examples of dynamic intersections

Multi-Agent Deep Reinforcement Learning to Manage Connected Autonomous Vehicles at Tomorrow's Intersections

This work presents Reinforced Autonomous Intersection Management (RAIM), a centralized Multi-Agent Deep Reinforcement Learning (MADRL) system that "periodically controls the speed of CAVs passing through an intersection" [3]. The system uses Twin Delayed Deep Deterministic Policy Gradient (TD3) algorithm with a centralized critic. Their state space design provides valuable reference for defining vehicle observations including position, speed, angle, lane, and turn intention. They achieve "reduction in travel time by 59% and time loss by 95%" in simulation. However, architecture relies on a central controller, creating a single point of failure.

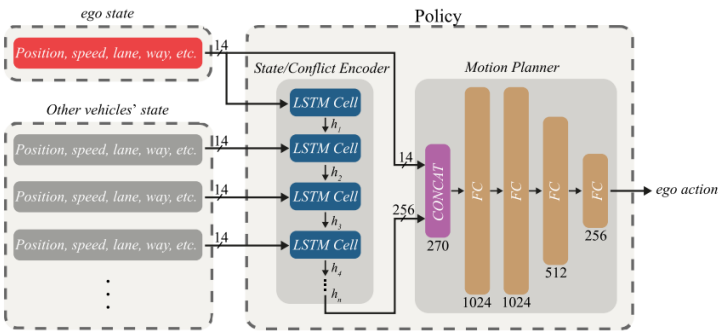


Figure 4 - The centralized RAIM architecture where one controller manages all vehicles

GamePlan: Game-Theoretic Multi-Agent Planning with Human Drivers at Intersections, Roundabouts, and Merging

This paper uses game theory to develop an auction where a vehicle's "bid" is its driving style (aggressive vs. conservative) [4]. The author states: "Vehicle place a single bid to get the right-of-way." The outcome is an optimal turn-based ordering where more aggressive drivers go first. They use CMetric algorithm to classify drivers as aggressive or conservative. While this captures driver's behavior, it treats negotiation as a one-time decision rather than a continuous process. They also note that their approach does not guarantee optimality for human drivers with varying social preferences.

Risk-Aware Reinforcement Learning for Cooperative Autonomous Vehicle Coordination with Adaptive Risk Sensitivity and Multi-Agent Optimization

This recent paper proposes Risk-Aware Reinforcement Learning (RARL) which "couples adaptive risk sensitivity with Bayesian risk estimation in a cooperative multi-agent setting" [5]. Their framework achieves "up to 30% fewer collisions, about 25% higher throughput, and roughly 20% faster training convergence." They use INTERACTION dataset for validation, proving its usefulness for intersection research. They also demonstrate that explicit risk modeling significantly enhances policy robustness.

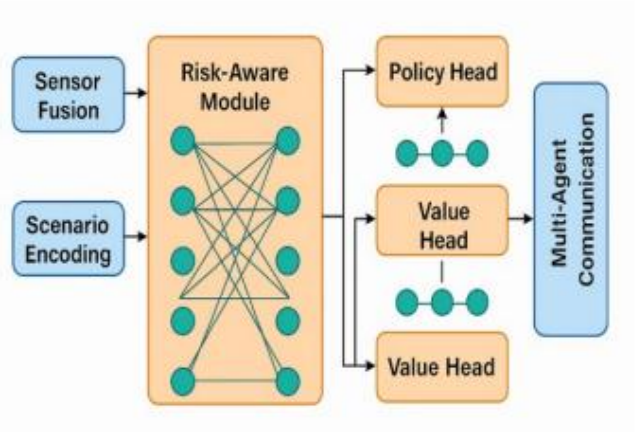


Figure 5 - The proposed RARL architecture

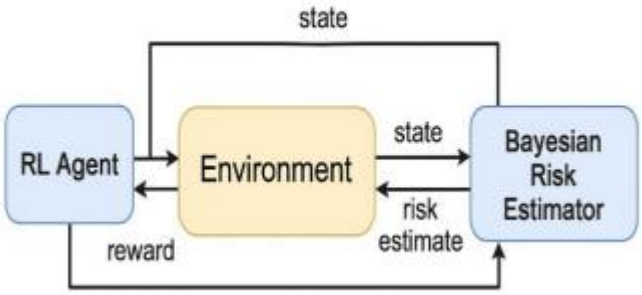


Figure 6 - Illustrates the control algorithm

Right-of-way based multi-agent deep reinforcement learning for collaborative decision-making at unsignalized intersection

This is the most directly relevant recent work. The authors propose a "centralized right-of-way based decision-making framework under Vehicle-Road Cooperative System (VRCS)" with a "right-of-way cooperator (RWC)" that "greatly reduces conflicts among vehicles" [6]. They achieve zero collision rate - but critically, this is achieved by adding external rules. They admit: "MADRL-based scheme cannot make CAVs fully abide by the right-of-way rules through RL training alone." They train for 1500 episodes using Multi-Agent Proximal Policy Optimization (MAPPO) with parameter sharing. This proves that pure learning without rules is still an open problem - exactly the gap this research addresses.

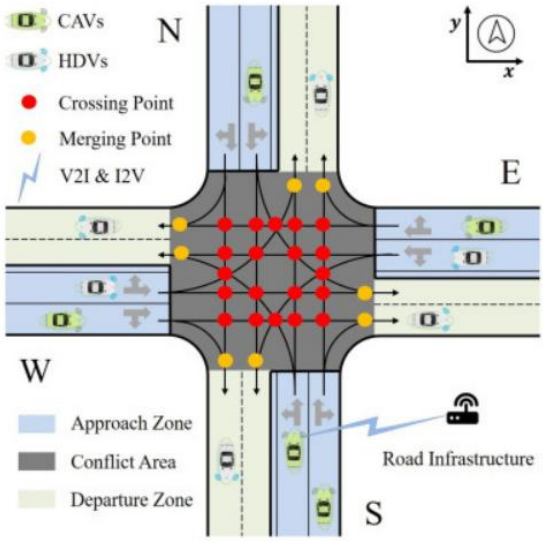


Figure 7 - The conflict points at a four-way unsignalized intersection, with red dots for crossing conflicts and yellow dots for merging conflicts

How similar or different are automated vehicle and human-driven vehicle crash patterns? Findings from crash sequence analysis

This recent study analyzed 555 autonomous vehicle crashes and 39,270 human-driven vehicle crashes [7]. Key finding: " Autonomous vehicle crashes at intersections occur nearly twice as often as HDVs (68.5% vs. 34.9%)." This provides strong evidence that intersections are a critical challenge for autonomous vehicles specifically, not just a general traffic problem. The study used crash data from multiple sources including the Department of Motor Vehicles (DMV).

Adaptive game-theoretic decision-making with driving style recognition for autonomous vehicles in uninterrupted traffic flows at intersections

This paper proposes a game-theoretic decision-making framework that explicitly integrates driving style recognition [9]. The authors develop "a hierarchical game-theoretic architecture integrating kinematic state evolution, feasibility constraints, and interactive behavior modeling" for multi-vehicle interactions. Their key contribution is an "online adaptive driving style estimation system that dynamically infers heterogeneous driver behaviors via real-time kinematic feature extraction and random forest regression." The framework demonstrates "enhanced interaction adaptability (more than 10% efficiency improvements) compared to conventional non-adaptive methods."

Cooperative Driving at Multiple Unsignalized Intersections in Fully Autonomous Driving Scenarios

This research addresses the limitation of single-intersection optimization by modeling "the coordination of multiple unsignalized intersections as a multi-agent sequential decision problem" solved through a two-stage method [10]. They use "multi-agent proximal policy optimization (MAPPO) algorithm to learn the global priorities" and "a hybrid filtered beam search [to] determine safe and efficient CAV passing orders." Their method outperforms state-of-the-art approaches in minimizing congestion.

Decision-Making of automated vehicles under diverse risky pedestrian crossing behaviors

This paper addresses pedestrian-related risks at uncontrolled midblocks [12]. They develop "a high-fidelity multi-agent simulation platform [that] replicates the dynamic interactions among AVs, human-driven vehicles, and pedestrians" and propose "a deep reinforcement learning (DRL)–based decision-making framework [that] integrates risk assessment with safety filtering." Their approach "substantially improves safety margins and control smoothness in complex scenarios."

Existing Products/Services

Current commercial systems from Waymo, Cruise, and Tesla handle intersections using High-Definition (HD) maps and rule-based logic. Waymo's published safety data shows lower crash rates than humans overall, but they still experience disengagements at intersections. These systems do not perform explicit multi-agent negotiation. Tesla's Full Self-Driving (FSD) has documented issues with unprotected left turns at intersections.

Highlight limitations of existing systems

Paper	What They Did	Missing Element
Autonomous Driving at Unsignalized Intersections: A Review of Decision-Making Challenges and Reinforcement Learning-Based Solutions [1]	Reviewed existing approaches and challenges in autonomous intersection decision-making.	No new solution proposed; it only identifies problems without implementing a practical system.
Cyber Traffic Light: Safe Cooperation for Autonomous Vehicles at Dynamic Intersections [2]	Propose a temporary central manager using V2V communication to coordinate vehicles.	Rely on centralized control instead of direct negotiation between vehicles.
Multi-Agent Deep Reinforcement Learning to Manage Connected Autonomous Vehicles at Tomorrow's Intersections [3]	Introduced a centralized MARL system (RAIM) where one controller manages all vehicles.	Creates a single point of failure if the central controller fails.
GamePlan: Game-Theoretic Multi-Agent Planning with Human Drivers at Intersections, Roundabouts, and Merging [4]	Uses game theory where vehicles make a one-time bid based on driving style.	Negotiation is not continuous; vehicles make a single decision without further interaction.
Risk-Aware Reinforcement Learning for Cooperative Autonomous Vehicle Coordination with Adaptive Risk Sensitivity and Multi-Agent Optimization [5]	Applies risk-aware reinforcement learning using Bayesian risk estimation.	Focuses mainly on safety modeling rather than negotiation between vehicles.
Right-of-way based Multi-Agent Deep Reinforcement Learning for Collaborative Decision-Making at Unsignalized Intersection [6]	Combines MARL with external right-of-way rules using a right-of-way cooperator.	Depending on external rules, showing pure reinforcement learning alone is insufficient.
How Similar or Different Are Automated Vehicle and Human-Driven Vehicle Crash Patterns? Findings from Crash Sequence Analysis [7]	Analyzes crash patterns between autonomous vehicles and human-driven vehicles.	Does not propose a new solution; only analyzes accident data.
Adaptive Game-Theoretic Decision-Making with Driving Style Recognition for Autonomous Vehicles in Uninterrupted Traffic Flows at Intersections [9]	Uses game theory with driving style classification for decision making.	Driving style is identified once and not updated during interaction.

Cooperative Driving at Multiple Unsignalized Intersections in Fully Autonomous Driving Scenarios [10]	Uses MAPPO to coordinate vehicles across multiple intersections.	Still depends on centralized coordination rather than decentralized negotiation.
Decision-Making of Automated Vehicles under Diverse Risky Pedestrian Crossing Behaviors [12]	Uses DRL to manage interactions between vehicles and pedestrians.	Focuses on pedestrian behavior rather than vehicle-to-vehicle negotiation.

Table 4 - Highlight limitations of existing systems

1.3 Research Gap

After reviewing the existing literature, current intersection management systems for autonomous vehicles still have significant limitations. Although many studies have proposed solutions using reinforcement learning, game theory, or centralized traffic management, none of them enable **fully decentralized and continuous negotiation between vehicles**.

Most existing approaches rely on **centralized control**, where a single system coordinates all vehicles at the intersection. For example, some reinforcement learning frameworks use a central controller to manage vehicle decisions and traffic flow [3][10]. While these systems can improve efficiency, they create a **single point of failure** and are difficult to scale in real-world environments. Similarly, some studies introduce temporary traffic managers that coordinate vehicles through communication protocols, but these still rely on centralized decision-making rather than true peer-to-peer interaction [2].

Other research applies **game-theoretic approaches or driving style classification** to determine right-of-way decisions. However, these methods typically rely on **one-time decisions**, such as a single bid or a fixed driving style classification, which does not reflect how real drivers continuously adjust their behavior during interactions [4][9]. In addition, several reinforcement learning approaches require **external safety rules** to ensure safe coordination between vehicles. For example, Feng et al. [6] introduced a right-of-way cooperative mechanism to avoid conflicts but acknowledged that reinforcement learning alone could not guarantee rule-compliant behavior without additional rules.

Some studies focus mainly on **predicting vehicle behavior or modeling risk** rather than enabling active negotiation between vehicles [1][5]. Other work primarily analyzes crash patterns without proposing new solutions to the problem [7]. There are also studies that focus on pedestrian interactions rather than vehicle-to-vehicle coordination at intersections [12]. As a result, the fundamental problem of **how autonomous vehicles can dynamically negotiate right-of-way with each other in real time remains largely unresolved**.

The urgency of this problem is supported by recent crash analysis research. A study by Andriola et al. [7] found that **autonomous vehicles are nearly twice as likely to crash at intersections compared to human-driven vehicles**, highlighting the need for improved coordination and decision-making mechanisms. Although recent work has explored driving style recognition [9] and multi-agent reinforcement learning for intersection coordination [10], these systems still rely on centralized control or fixed decision strategies.

Therefore, a clear research gap exists **there is currently no system that enables decentralized, continuous negotiation among autonomous vehicles without relying on centralized controllers or external safety rules**.

To address this gap, this research proposes a **multi-agent reinforcement learning framework with explicit negotiation signals**, where each vehicle acts as an independent agent and communicates its negotiation intent in real time. Through this approach, vehicles can dynamically adjust their behavior and cooperatively determine right-of-way decisions at complex intersections without centralized coordination.

2. OBJECTIVE

2.1 Main Objective

The main objective of this research is to design, develop, and evaluate a decentralized Multi-Agent Reinforcement Learning (MARL) framework that enables autonomous vehicles to negotiate right-of-way at unsignalized intersections using explicit priority intent communication signals, with the goal of reducing collisions and improving traffic efficiency compared to traditional rule-based methods.

2.2 Specific Objectives

1. **To design a decentralized negotiation framework** in which autonomous vehicles communicate their driving intentions using continuous negotiation signals ranging from 0 to 1, inspired by driving style research [9].
2. **To implement a Multi-Agent Reinforcement Learning model using the MAPPO algorithm**, integrated with a negotiation layer and an LSTM-based observation encoder to process vehicle interaction data [1][6][10].
3. **To train autonomous vehicle agents in a simulated intersection environment** using traffic simulators such as SUMO and CARLA, enabling the agents to learn cooperative right-of-way negotiation through reinforcement learning [6].
4. **To evaluate the proposed system using real-world traffic datasets**, including INTERACTION, NGSIM, and HighD datasets, in order to measure safety, traffic flow efficiency, and negotiation performance [1][5].
5. **To compare the proposed negotiation-based system with existing rule-based and centralized intersection management approaches**, to determine improvements in collision reduction and traffic throughput.
6. **To develop and document a functional prototype of the decentralized negotiation system** that demonstrates the feasibility of real-time multi-agent coordination at complex intersections.

3. METHODOLOGY

3.1 Research Approach

This research uses Design Science Research combined with experimental validation.

- Design Science: We create a new system, a decentralized negotiation framework for autonomous vehicles at intersections.
- Experimental validation: The system is tested in simulated environments to evaluate safety, efficiency, and negotiation performance.
  This approach is standard in computer science and engineering research [1][3][6].

3.2 Technical Methods and Algorithms

Step 1: Problem Formulation

The problem is modeled as a Decentralized Partially Observable Markov Decision Process (Dec-POMDP).

- Each vehicle is an agent with partial knowledge of other vehicles [1].

Step 2: State Space (What Each Vehicle Sees)

Every 0.1 seconds, each vehicle collects:

Category	Information	Format
Ego State	Speed, Position (x,y), Heading, Lane, Turn Intention, Distance to stop line	m/s, meters, degrees
Neighbor Input	Up to 5 vehicles: relative positions, speeds, neighbour_signal (received via V2V)	Δx, Δy meters, m/s, 0-1
Environment	Traffic density	low / medium / high

Table 5 - State Space (What Each Vehicle Sees)

Step 3: Action Space (What Each Vehicle Does)

Output	Range	Purpose
Acceleration	-3 to +2 m/s²	Control speed (brake or accelerate)
priority_intent	0–1	Communicate assertiveness to other vehicles (novel feature)

Table 6 - Action Space (What Each Vehicle Does)

The priority_intent signal meaning:

- 0.0 - 0.3: "I am yielding, you go first" (maps to deceleration behavior)
- 0.4 - 0.6: "I am neutral, still deciding" (maps to maintaining speed)
- 0.7 - 1.0: "I am asserting my right-of-way" (maps to acceleration behavior)

This three-point scale is supported by recent work on driving style recognition [9] and the spectrum from conservative to aggressive drivers [4].

Step 4: Neighbour Encoder (LSTM)

- Handles variable numbers of vehicles.
- Converts inputs from 2–5 neighbours into a fixed-size vector (128 numbers) for the neural network [1].
- Inputs per neighbour: relative position, speed, neighbour_signal.

Step 5: MAPPO Algorithm

Multi-Agent Proximal Policy Optimization (MAPPO) is used for learning [6][10]:

- Actor (Driver): Decides acceleration and priority_intent based on vehicle observations.
- Critic (Judge): Evaluates all vehicles' actions during training to improve policy.
- CTDE (Centralized Training, Decentralized Execution):
    - Training: vehicles share information in simulation.
    - Execution: each vehicle acts independently [6].

Step 6: Reward Function

Vehicles learn using rewards (+) and penalties (-) [3][5][12]:

Situation	Points	Reason
Collision	-1000	Avoid accidents
Successfully cross	+100	Goal achieved
Waiting each second	-0.1	Avoid delays
Driving fast	+0.1 × (speed / limit)	Efficient travel
Good cooperation	+10	Successful negotiation
Deadlock	-5	Negotiation failed
Breaking right-of-way	-10	Follow traffic rules
Appropriate yielding	+5	Good social behavior

Table 7 - Reward Function

Step 7: Training Process

Curriculum learning is applied [3][6]:

Phase	Traffic Density	Episodes	Goal
1	Low (200 veh/h)	500	Learn basic negotiation
2	Medium (600 veh/h)	500	Handle more vehicles
3	High (1200 veh/h)	500	Manage dense traffic

Table 8 - Training Process

Total episodes: 1500 (similar to "Right-of-way based multi-agent deep reinforcement learning for collaborative decision-making at unsignalized intersection" research paper [6]).

3.3 Tools and Platforms

Tool	Purpose	Why Chosen
SUMO	Traffic simulation	Free, widely used in literature [2][3][4][6][10]
CARLA	High-fidelity simulation	Realistic vehicle interactions [1][5][12]
PyTorch	Neural network implementation	Industry standard
Ray/RLlib	MARL training	Supports MAPPO
Python	Programming	Standard, widely supported
NumPy / Pandas	Data processing	Efficient for large datasets

Table 9 - Tools and Platforms

3.4 Data Collection and Ethical Considerations

Datasets used:

Dataset	Contents	Use
INTERACTION [5]	Real intersection negotiations	Validation against human behavior
NGSIM [1]	Vehicle trajectories (11M+ rows)	Calibrate vehicle motion
HighD [1]	Highway trajectories (25 Hz)	Calibrate speed and acceleration
SUMO	Synthetic traffic data	Train millions of episodes

Table 10 - Datasets used

Ethical considerations:

- All datasets are public and anonymized.
- No human subjects are involved.
- Research is conducted entirely in simulation.
- Safety monitors will be used in future real-world testing.
- Addresses the critical safety issue: AVs crash at intersections almost twice as often as humans [7].

3.5 Validation Strategy and Performance Metrics

Baselines for Comparison:

- FCFS (First-Come First-Served) rule-based [2]
- Centralized MARL (RAIM) [3]
- MARL + Rules (RWC) [6]
- Game-theory one-time bid [4]
- Centralized MAPPO [10]

Metrics for Evaluation:

Metric	Meaning	Target
Collision Rate	% of episodes with crashes	0%
Average Travel Time	Time to cross intersection	<15 s
Throughput	Vehicles per hour	+25% vs rule-based [5]
Post-Encroachment Time (PET)	Safety gap between vehicles	>2 s
Deadlock Frequency	% of episodes with all vehicles stuck >3 s	<1%
Fairness Index	Variance in waiting times	>0.8

Table 11 - Metrics for Evaluation

4. HIGH-LEVEL SYSTEM ARCHITECTURE

Overview of Proposed Solution

The proposed system is a **multi-agent negotiation framework for autonomous vehicles at unsignalized intersections**. Each vehicle acts as an independent agent that observes its own state, nearby vehicles, and traffic conditions every 0.1 seconds. Information about neighboring vehicles is processed using an **LSTM encoder**, which helps handle a changing number of vehicles around the intersection. These features are then passed to a **decision network trained using the MAPPO algorithm**, which generates two outputs: the vehicle's **acceleration** and a **priority_intent signal** used to communicate negotiation intentions to other vehicles. The outputs are shared with nearby vehicles via V2V communication, forming a **real-time negotiation loop** where each vehicle's actions influence others. This process repeats until all vehicles safely clear the intersection.

System Diagrams

Overall system Diagram

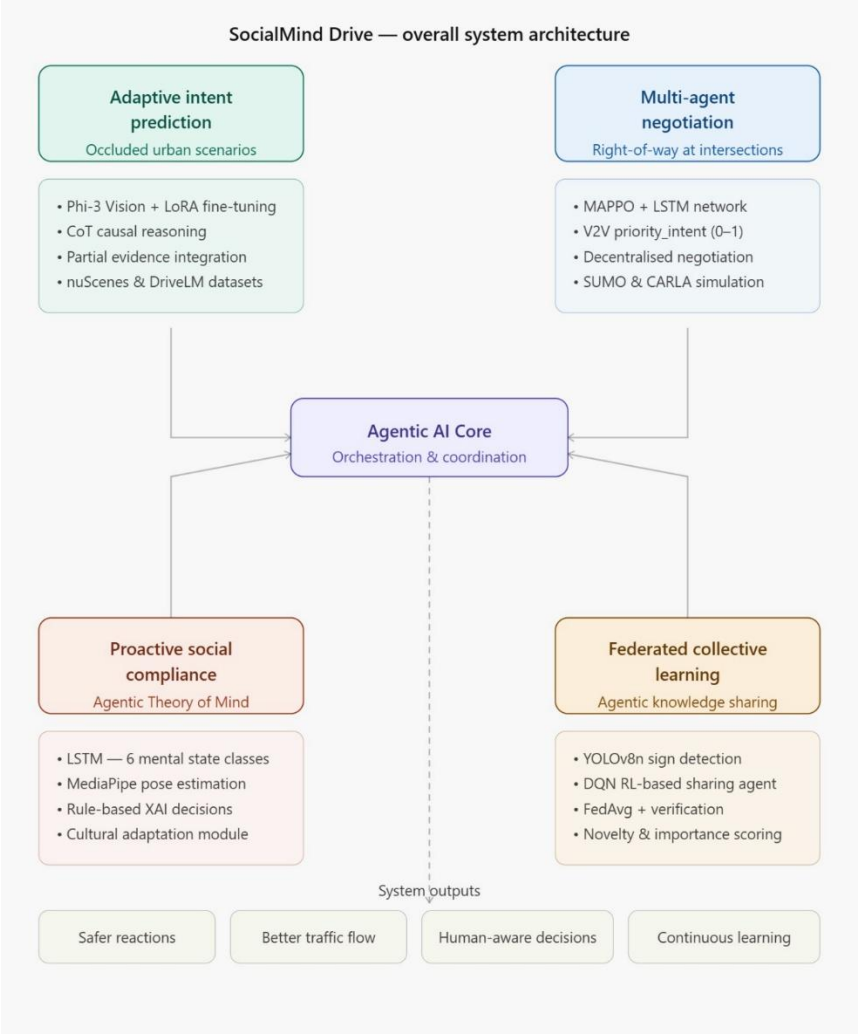


Figure 8 - Overview of Proposed Solution

Individual Component System Diagram

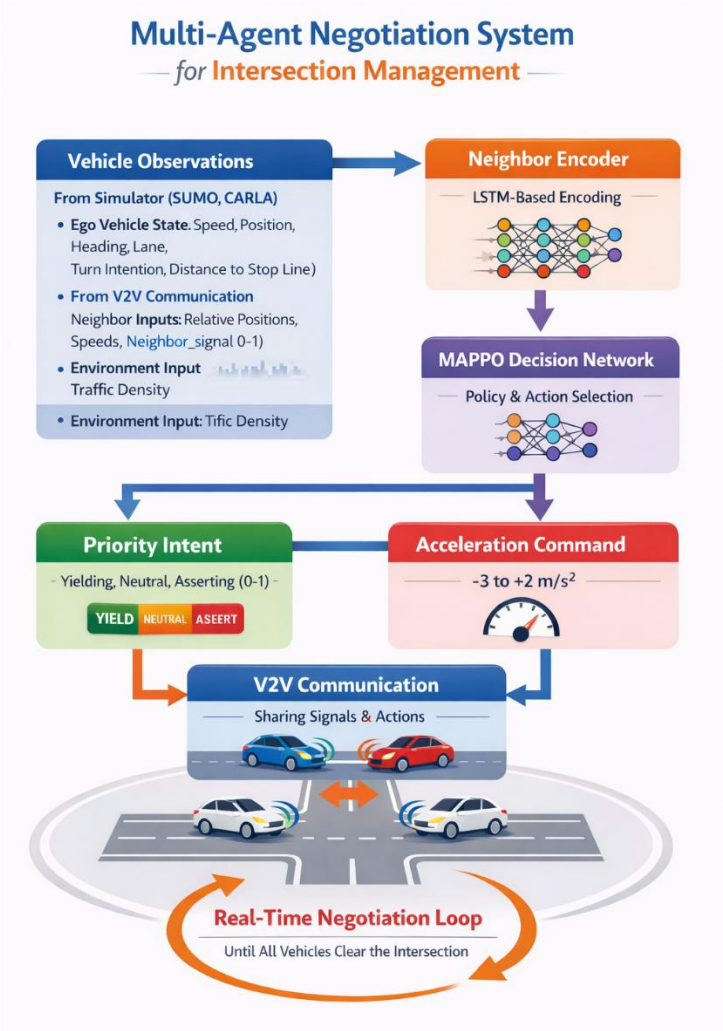


Figure 9 - Component Detailed Architecture

Data Flow Explanation

1. Simulation generates data (SUMO/CARLA) = Each car receives observations including neighbour_signal from others
2. Neighbour Encoder (LSTM) processes variable number of other vehicles = Fixed-size representation
3. Feature Fusion combines ego state, LSTM output, and environment
4. Decision Network (MAPPO Actor) processes features = Outputs Vehicle performs action and priority_intent
5. Vehicle Actions = Vehicle performs action based on decision: accelerate, slow down, stop or move forward

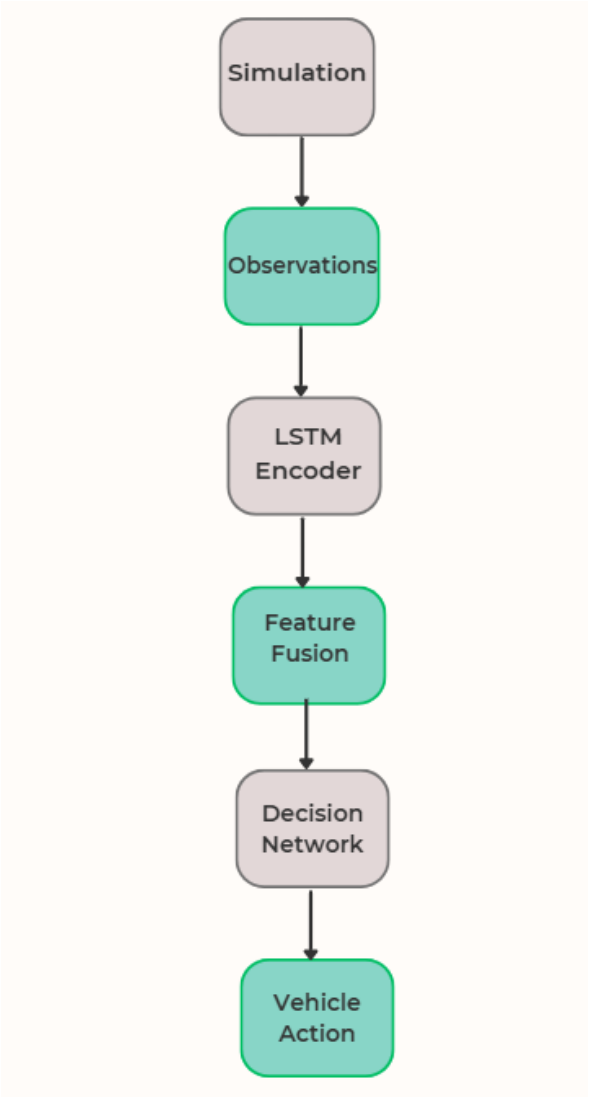


Figure 10 - Data flow

Scalability and Security Considerations

Scalability:

- LSTM handles any number of nearby vehicles (tested up to 5, works for more) [1]
- Parameter sharing means policy works for any number of agents [6]
- Decentralized execution means no central bottleneck
- Can scale from 2 to 20+ vehicles in the control zone

Security:

- All datasets are public and anonymized
- Simulation only during development (no real-world risks)
- Communication authentication will be considered for future real-world deployment
- Graceful degradation if communication fails (fallback to implicit negotiation)
- Safety monitors override unsafe decisions regardless of received signals

5. USER REQUIREMENTS

Stakeholder Identification

Stakeholder	Role	Needs
Autonomous Vehicle Manufacturers	System integrators	Reliable intersection negotiation and safety guarantees
Autonomous Vehicle Software Developers	System implementers	Well-documented APIs and modular system design
Passengers	End users	Safe travel, comfort, and timely arrival
Drivers of Human Vehicles	Road users interacting with AVs	Predictable AV behavior and safe interaction
Pedestrians and Cyclists	Vulnerable road users	Clear and predictable vehicle behavior
Regulators (e.g., National Highway Traffic Safety Administration, Department of Motor Vehicles)	Certification and regulatory bodies	Verifiable safety performance and rule compliance
Insurance Companies	Risk assessment organizations	Lower crash rates and reduced insurance claims

Table 12 - Stakeholder Identification

Functional Requirements

Requirement	Priority
The system shall enable vehicles to negotiate with 2–10 other vehicles simultaneously using *priority_intent* signals	High
The system shall process neighbour_signal inputs from nearby vehicles at 10 Hz frequency	High
The system shall generate acceleration commands at 10 Hz frequency	High
The system shall achieve zero collisions in simulation testing	Critical
The system shall support explicit priority_intent signals (0–1) for vehicle communication	High
The system shall support vehicles with different turning intentions (left, straight, right)	Medium
The system shall adapt to different traffic densities (200–1200 vehicles per hour)	High
The system shall include safety monitors that override unsafe decisions	Critical

Table 13 - Functional Requirements

Non-Functional Requirements

Requirement	Target
Inference latency (decision time)	< 50 ms per decision
Training duration	1500 episodes (approx. 4 months training process)
Collision rate in test scenarios	0%
Throughput improvement	≥ 25% compared to FCFS method
Model/policy size for deployment	< 100 MB
Communication bandwidth per vehicle	< 1.5 MB/s

Table 14 - Non-Functional Requirements

6. COMMERCIALIZATION PLAN

Target Market and Customer Persona

Primary market: Autonomous vehicle(AV) manufacturers developing Level 4/5 systems.

Customer persona: Director of Autonomous Driving at automotive OEM or AV startup (companies like Tesla, Waymo, Cruise, Ford, GM, BMW, Mercedes).

Market size: Global autonomous vehicle market projected to exceed $2 trillion by 2030. With 88% of AV crashes occurring at intersections [1] and AVs crashing twice as often as humans at intersections [7], there is urgent need for better intersection solutions.

Value Proposition

Problem	Our Solution	Benefit
AVs crash twice as often as humans at intersections [7]	Zero-collision negotiation with priority_intent signals	Saves lives, reduces liability
88% of AV crashes occur at intersections [1]	Targeted intersection solution using neighbour_signal processing	Addresses biggest risk area
Rule-based systems fail in new situations [2]	Learned negotiation adapts	Works everywhere
Centralized systems have single point of failure [3][10]	Decentralized decision-making	More reliable
Current MARL needs external rules [6]	Safety emerges from learning with explicit signals	Truly autonomous

Table 15 - Value Proposition

Revenue Model

Revenue Stream	Description	Timeline
Software licensing	Per-vehicle license fee	Year 2+
Development partnerships	Co-development with OEMs	Year 1-3
Consulting services	Integration support	Year 1-3

Table 16 - Revenue Model

Cost Estimation

Category	Item	Cost (USD)
Personnel	Lead Researcher (4 months)	$20,000
Personnel	Assistant (4 months)	$15,000
Cloud	GPU instances for training (1500 episodes)	$3,000
Cloud	Storage (datasets, models)	$500
Publication	Conference fees	$500
Contingency	10% of total	$3,900
Total		$42,900

Table 17 - Cost Estimation

Pricing Strategy

License Type	Price	Includes
Evaluation	$10,000/year	Limited testing, email support
Development	$50,000/year	Full access, priority support
Production	$20/vehicle	Per-vehicle royalty

Table 18 - Pricing Strategy

Example: 100,000 vehicles × $20 = $2,000,000 revenues

Competitive Advantage

Competitor	Limitation	Our Advantage
Rule-based systems (FCFS) [2]	Brittle, cannot handle edge cases	Learned policies adapt using priority_intent
Centralized MARL (RAIM) [3]	Single point of failure	Truly decentralized with neighbour_signal processing
MARL + Rules (RWC) [6]	Safety from rules, not learning	Safety emerges from learning with explicit signals
Game theory (one-time bids) [4][9]	One-time decisions only	Continuous negotiation with 0-1 signals
Centralized MAPPO [10]	Centralized learning	Decentralized execution with V2V communication
Current AV systems (Tesla, Waymo)	No explicit negotiation	First-to-market with priority_intent negotiation capability

Table 19 - Competitive Advantage

Intellectual Property Considerations

- **Patents:** priority_intent signal protocol, neighbour_signal processing architecture
- **Copyright:** Source code, trained models
- **Licensing:** Open-source core algorithms, proprietary optimizations

7. BUDGET AND JUSTIFICATION

Category	Item	Cost (USD)	Justification
Personnel	Lead Researcher (4 months)	$20,000	Algorithm design, implementation, writing
Personnel	Assistant Researcher (4 months)	$15,000	Simulation setup, data analysis
Cloud	AWS/GCP GPU instances (1500 episodes)	$3,000	MAPPO training requires significant compute [6]
Cloud	Storage (INTERACTION, NGSIM, HighD datasets, models)	$500	Store and process large datasets
Software	Development tools	$0	All open-source (SUMO, CARLA, PyTorch, etc.)
Publication	Conference fees (IEEE/ITSC)	$500	Disseminate results
Contingency	10% of total	$3,900	Unexpected costs
TOTAL		**$42,900**	

Table 20 - Budget and Justification

8. WORK BREAKDOWN STRUCTURE (WBS)

WBS	Task	Deliverable	Hours	Week
1.1	Literature Review	15+ papers reviewed	40	1–2
1.2	Gap Analysis	Gap statement + novelty points	15	2
1.3	Methodology Design	Complete methodology section	20	3
1.4	Architecture Design	Architecture + data flow diagrams	15	4
1.5	Proposal Writing	Complete proposal document	10	4
2.1	Simulation Setup	SUMO + CARLA environments ready	35	5
2.2	Dataset Preparation	INTERACTION, NGSIM, HighD processed	40	6–7
2.3	TraCI Interface	SUMO-Python communication working	20	7–8
2.4	Calibration	Simulation parameters calibrated	20	8
3.1	LSTM Encoder	Implemented neighbour encoder	30	8–9
3.2	Actor–Critic Networks	Decision networks implemented	35	9
3.3	MAPPO Implementation	MARL training pipeline ready	40	10–11
3.4	Reward & Communication	Reward function + V2V module	30	11
3.5	System Integration	Integrated model with simulation	30	12
4.1	Model Training	1500 episodes completed	140	13–16
5.1	Validation & Testing	Metrics + comparison results	100	15–18
6.1	System Integration	Integrated with other components	60	18–20
6.2	Documentation	Final report + research paper	40	21–22
6.3	Final Presentation	Final presentation and demo	20	22

Table 21 - Work Breakdown Structure

9. GANTT CHART

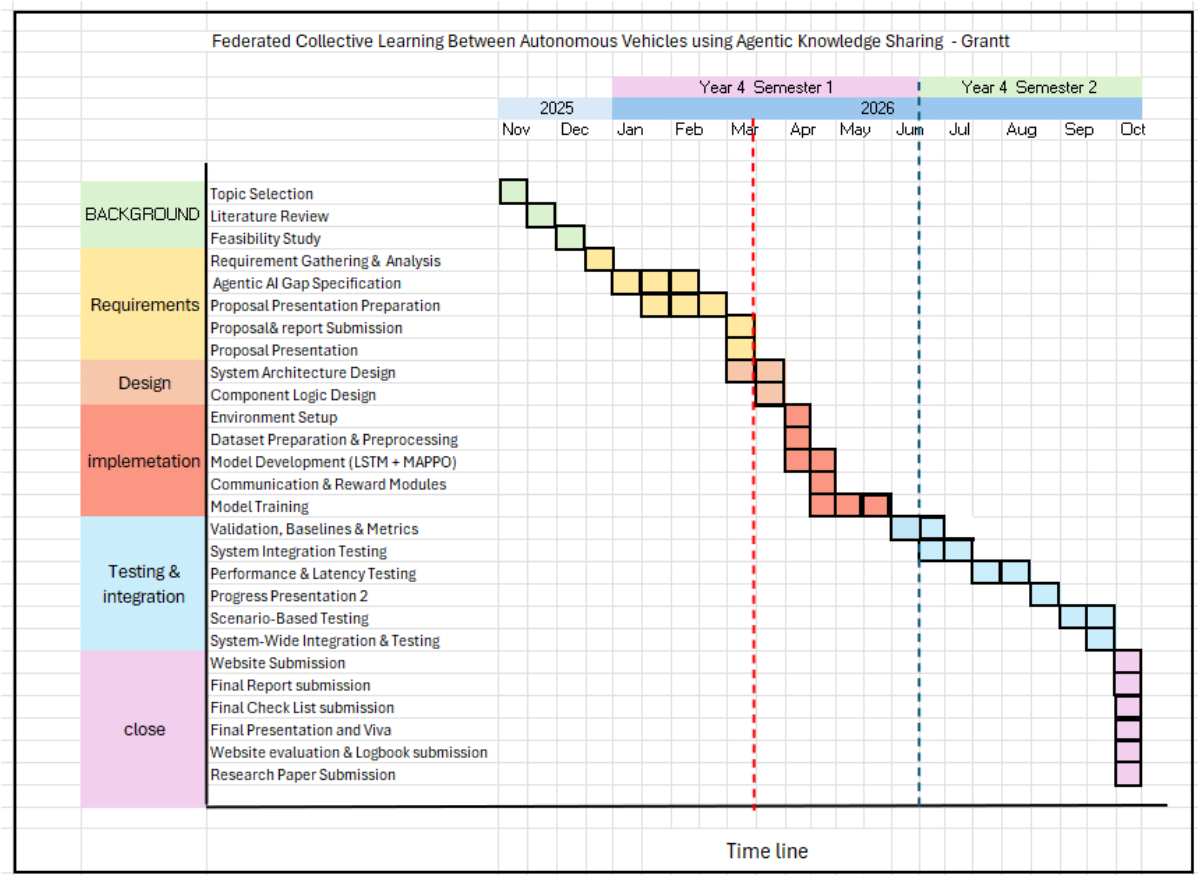


Figure 11 - Gantt Chart

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